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Generative AI Meets Knowledge Management: Insights From Software Development Practices

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ABSTRACT

Recent developments in generative artificial intelligence (GenAI) have raised the interest of knowledge management (KM) scholars in artificial intelligence. By harnessing GenAI, KM processes become more efficient, scalable and adaptable to the needs of organisations and users. Taking the theoretical lens of the Technology–Organisation–Environment (TOE) framework for technology adoption, this study investigates the factors influencing the adoption of GenAI among software developers, as a specific group of knowledge workers, in knowledge creation and sharing. Our interviews with 11 developers from 8 countries were analysed by combining an inductive and a deductive approach. We identified five themes alongside the TOE framework, emphasising the adoption factors of GenAI in software development, particularly within KM. Based on our findings, we discuss how GenAI is adopted for KM in software development. In particular, the interviewees liked the GenAI affordances of solving simpler programming tasks efficiently and rapidly. However, GenAI requires expertise to review and modify the code, write suitable prompts and evaluate the reliability of the provided output. Knowledge exchange with fellow programmers is partly, but not entirely, replaced by exchange with GenAI as a new development team member, as it is more efficient. Nevertheless, continuous learning, adaptation and ethical consideration are needed to realise the full benefits of GenAI tools in software development. This study's findings provide broad practical insights into addressing challenges stemming from GenAI integration in KM processes. Additionally, it prompts a critical reflection on the necessity of revising existing KM theoretical models in light of the emergence of AI-generated knowledge.

1 | Introduction

With advancements in technology, especially artificial intelligence (AI), the strategies and tools used in knowledge management (KM) are evolving (Wang et al. 2022; Jarrahi et al. 2023). AI has various applications in KM, offering benefits such as real-time knowledge sharing, for example, providing real-time feedback to boost the effectiveness of digital collaboration (Porter and Grippa 2020), improving knowledge discovery like using deep learning techniques for document classification on a large scale (Jiang et al. 2022), and enhancing knowledge delivery like adaptive learning and personalising knowledge distribution (Almohammadi and Hagras 2013). According to Jarrahi et al. (2023), AI can support fundamental processes of KM, including the creation, storage, retrieval, sharing and application of knowledge. Studies such as those by Li et al. (2018) on the application of AI in KM and the work by Davenport and Ronanki (2018) on the broader impact of AI on business processes provide valuable insights into the multifaceted advantages of AI. To sum up, it is a widespread

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opinion among scholars (Mohammad et al. 2022; Taherdoost and Madanchian 2023) and practitioners (Accenture 2023) that AI is offering new foundations to transform KM (Jarrahi et al. 2023), as AI and KM at their core are about knowledge (Rhem 2017). Although KM is concerned with managing knowledge in organisations, AI is focused on developing systems that mimic human knowledge and learning activities (Jarrahi et al. 2023).

Although AI has the potential to support KM in many ways, research on the relationships between AI and KM is still scarce (Alghanemi and Al Mubarak 2022; Al Mansoori et al. 2021), and there is little empirical evidence regarding integrating AI technologies into KM processes. Extant studies are primarily conceptual, based on the analysis of existing literature, and consider the entire set of AI technologies. Furthermore, they underline the advantages and benefits of AI while devoting less attention to the challenges and problems raised by its adoption in real-life cases.

How new technologies will enable the re-design and re-organisation of knowledge work is an important future research topic (Sarka et al. 2019). It is a common opinion that the recent advances in AI, especially the public availability of generative AI (GenAI), such as GPT, Dall-E or Copilot, will change how we work and communicate. GenAI encompasses computational technologies that create new and meaningful content based on training data, such as text, images or audio (Feuerriegel et al. 2024). The development of the last version of ChatGPT, a very popular GenAI model, has raised great interest among scholars and practitioners regarding its possible use in organisational management (Ayinde et al. 2023; Chui et al. 2023) and also in KM (Alavi et al. 2024).

Our paper aims to provide insights into how AI can affect KM processes in knowledge-intensive organisations. In particular, it examines how software developers use GenAI to generate new knowledge, especially new software code, and share knowledge regarding development problems. Software developers are a specific group of knowledge workers who combine creativity in developing new applications (knowledge creation) and rational programming rules (explicit knowledge) with their expertise and experience (tacit knowledge) (Aurum et al. 2008). They usually work in development teams and thus regularly share knowledge (Khalil and Khalil 2020). While several authors have already published about using GenAI in software development (e.g., Rajbhoj et al. 2024; Ulfsnes et al. 2024), this topic has yet to be analysed from a KM perspective. To address this research gap, we interviewed 11 software developers in eight countries about their experiences using GenAI daily and how it impacts knowledge creation and sharing. Furthermore, in line with Wael AL-khatib (2023) and Maragno et al. (2023), we employed the Technology–Organisation–Environment (TOE) framework of technology adoption (Tornatzky et al. 1990) as an additional theoretical lens to interpret the empirical findings.

The paper is structured as follows: Section 2 provides the background of our study regarding work in KM and GenAI, the adoption of GenAI in software development and the TOE framework for adopting new technologies. Section 3 illustrates our qualitative research approach. Section 4 presents the study's findings, which we discuss in Section 5. Section 6 provides conclusive remarks and practical implications.

2 | Background of the Study

2.1 | KM and GenAI

2.1.1 | The Intersection of KM and AI

A strict link between KM and AI was highlighted more than two decades ago by Liebowitz (2001), who considered AI to be one of the critical blocks for developing and advancing the field of KM. However, due to the limitations of previous AI technologies, KM scholars and practitioners have generally overlooked such a linkage. In recent years, the opinion is rapidly diffused that the latest advances in AI cannot only provide new foundations for profoundly transforming KM in organisations (Jarrahi et al. 2023; Sumbal and Amber 2025) but also that adopting a KM perspective can enable a better implementation of AI (Iaia et al. 2024). Hence, the number of studies on the relationships between AI and KM is growing rapidly (Nakash and Bolisani 2024). Edwards and Lönnqvist (2023), in their paper on the future of KM, underline the need to go deeper into the connection between AI and KM, asserting that 'AI desperately needs KM' to exploit its opportunities fully.

GenAI, especially the availability of ChatGPT, has boosted the interest of KM scholars in AI. GenAI uses advanced algorithms called large language models (LLMs) based on deep learning and artificial neural networks to create new content like text, images and media (Chui et al. 2023; Soni 2023). These models can process huge and varied unstructured datasets and perform different tasks, such as classifying data, reading and writing texts, composing music and developing software code. GenAI can process natural language to help employees retrieve stored internal knowledge by phrasing questions like they might ask a human and engage in an ongoing dialogue (Chui et al. 2023). All the previous features, particularly its back-and-forth conversational mechanism (Hu et al. 2023), make GenAI the most considerable disruption in the KM field (Dwivedi et al. 2023). The potential impact of GenAI, especially ChatGPT, on KM has been the subject of recent academic research (Kaczorowska-Spychalska et al. 2024; Sumbal and Amber 2025).

Alavi et al. (2024) analysed the organisational implications of GenAI from a KM perspective. In particular, they investigated how GenAI impacts the different KM processes (creation, storage, transfer and application), emphasising the opportunities and challenges this technology raises. Regarding knowledge creation, the authors affirm that GenAI can facilitate the combination and internalisation modes of knowledge creation. As regards the combination mode, GenAI excels at generating new content, identifying patterns and making predictions, and in doing this, it serves as an active knowledge co-creator. Furthermore, GenAI supports internalisation through interactive, conversational engagement of users. GenAI can respond to user queries and provide coaching, contextual examples and in-depth explanations.

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Together with the opportunities, Alavi et al. (2024) identified six critical areas, as follows:

1. A loss of control over the knowledge generated by GenAI, because organisations have limited influence over the foundational knowledge the models acquire.
2. The risk is that an intense resort to GenAI may reduce human socialisation and knowledge collaboration.
3. The sidestepping of junior knowledge workers, whose traditional entry tasks of collecting information and drafting reports are now better and quicker performed by GenAI.
4. The lowering of knowledge internalisation, because some employees may accept GenAI-generated knowledge without internalising it for future use and development.
5. The possible dissuasion of employees from engaging in the externalisation of knowledge produced by GenAI.
6. The possible generation of inaccurate outputs (the so-called 'hallucinations').

These issues raise several research *questions* regarding the impact of GenAI on knowledge-sharing and collaboration among employees, its influence on the creation and maintenance of social networks, the assessment of relevance and reliability of the generated knowledge, and the effects of GenAI on internalisation and externalisation processes within organisations.

The integration of GenAI into KM brings up not only practical issues that are the primary focus of the present studies, but also theoretical considerations. Recently, scholars have questioned whether there is a need to refine existing theories or even develop new ones to move beyond the current hype surrounding GenAI (Böhm and Durst 2025; Brown et al. 2024).

2.1.2 | Empirical Studies on the Integration of KM and GenAI

This subsection reviews recent empirical studies that examined the implementation and impacts of GenAI within KM frameworks and perspectives. Such investigations are still few, and some of them analysed the effect of AI on individual knowledge workers and knowledge work rather than on KM as a management activity.

Woodruff et al. (2024) investigated how knowledge workers anticipate GenAI's effect on their industries in the future. Their findings suggest that the impacts of GenAI are context-dependent. Furthermore, whereas GenAI can automate routine tasks and serve as a valuable tool, it still requires human oversight and review for optimal results. This leads to several research questions on the intersection of GenAI and knowledge work concerning the need to adopt a human-in-the-loop approach (i.e., adding human review or oversight); the different expectations of knowledge workers about the impact of GenAI on their jobs; the unavoidable shifting of skill sets of existing and new professions; the dehumanisation of cognitive work, with the possible reduction of critical thinking; and the necessity of new accountability metrics.

Brynjolfsson et al. (2023) examined adopting a GenAI tool to provide conversational guidance for customer support agents. Their study shows that GenAI increases productivity for novice and low-skilled workers but minimally impacts experienced and highly skilled workers. In addition, GenAI contributes to disseminating the best practices of more able workers and helps newer workers move down the experience curve; improves customer sentiment; increases employee retention; and may be conducive to workers' learning. An interesting result concerns the large heterogeneity in effects across workers, which raises a research question about how to manage the top-performing workers, who contribute the most to training the AI system but see relatively few improvements in their productivity.

Ritala et al. (2024) analysed the use and utility of ChatGPT in knowledge work to understand how business value can be created from ChatGPT and similar GenAI tools. The authors introduce the term 'algorithmic assistance' to describe how ChatGPT is already transforming knowledge work in companies. The 'algorithmic' side highlights the sophisticated processes at the heart of GenAI, whereas the word 'assistance' underlines that ChatGPT can help humans perform—or avoid—tasks essential in knowledge work. In particular, the authors propose an interpretative framework based on the previous two dimensions that identifies the extent to which 'algorithmic assistance' from ChatGPT can be helpful to knowledge workers and how and to what extent it can or cannot replace human work. Their analysis confirms that knowledge workers acting in different application contexts, that is, marked by a particular type (more repetitive or more creative) of cognitive work, can be differently affected by GenAI.

Kaczorowska-Spychalska et al. (2024) provided insights into the integration of GenAI in KM practices and the opportunities and challenges of using GenAI in the context of knowledge work (knowledge creation, storage and retrieval, transfer and sharing and application). They found that when used to support the knowledge creation process, GenAI has the potential to create new declarative knowledge at a limited cost and can help people to be more creative, but, on the other hand, the knowledge it generates may be too complex to understand and lead to internal chaos. In conclusion, GenAI challenges the KM process, and potential advantages accompany threats. The authors claim that future research is needed to consider the advantages and disadvantages related to the use of technology in organisations separately and investigate the influence of GenAI on different stages of the KM process.

Sumbal et al. (2024) focused on the knowledge-creation process, primarily on how ChatGPT can facilitate KM, by taking as a conceptual reference Nonaka's SECI model (Nonaka 1994). Their findings show that ChatGPT can provide relevant support for all four conversion modes. This induced the authors to suggest adopting a ChatGPT-mediated KM system, as ChatGPT's advanced capabilities of natural language processing and high-quality text generation make it an ideal solution for knowledge flow and enhancing employee socialisation experience within organisations. It is worth noting that, unlike those seen so far, this study only considers the advantages offered by ChatGPT to support the knowledge-creating processes while neglecting the risks that this tool entails.

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In another work, Sumbal et al. (2024) explored how ChatGPT can enhance organizations' KM capability for improved decision-making and identified potential risks and opportunities. They developed a conceptual framework for integrating AI with KM. In doing this, the authors underline the opportunities and risks for integrating ChatGPT with KM and the potential benefits of a ChatGPT-mediated knowledge system. The former mainly concerns security risks and privacy concerns, a possible decrease in employees' critical thinking and innovative capabilities, and reduced employee interaction and collaboration. The latter results in better decision-making, personalised employee learning experiences, improved customer services and increased employee productivity. In conclusion, the authors suggest that future research should be devoted to investigating and exploring the phenomenon through practical dimensions of the ChatGPT-mediated KM system.

Retkowsky et al. (2024) used the theoretical lens of affordances to investigate the consequences for organisations when knowledge workers adopt and use ChatGPT. This implies analysing what ChatGPT allows employees to do in practice (Leonardi 2011), that is, how users employ the new technology (Gibson 1977). According to this approach, ChatGPT can be considered a malleable and multipurpose technology, given that its actual use largely depends on the context of the application. The authors identified six affordances: searching for information, structuring content, brainstorming ideas, writing a first draft, embellishing text and proofing work. While these affordances allow organisations to achieve productivity gains, they may have three side effects, both unexpected and long-lasting, that hit at an organisational scale: breaking of the existing knowledge ties because employees resort to ChatGPT instead of their colleagues; more limited managerial oversight of the quality of knowledge; and reconfiguration of work roles, as a result of tasks now partially or totally performed by ChatGPT.

The literature analysis confirms that empirical evidence on the adoption and use of GenAI as a KM tool is still very limited. However, from the first few studies, some indications can be drawn:

- The impacts of GenAI on KM are heterogeneous since they depend on the context (business sector, employee's role, nature of the cognitive work, etc.) where the new technology is used;
- The use of GenAI as a KM supporting tool provides several benefits but also raises some challenges and risks, so that GenAI can have both an enhancing and a hindering effect on organisational KM.
- The set of advantages and risks derived from integrating GenAI in KM is related to the specific KM activity for which the tool is used.

The above leads us to affirm that if generalist studies about the impact of GenAI on KM could be helpful in the initial phase, now there is a need to narrow and deepen the field of investigation, for example, by focusing on a specific application context.

2.2 | GenAI and Software Development

The adoption of GenAI affects software development by offering new opportunities, methodologies, and tools that significantly boost efficiency, minimise manual effort and foster creative programming practices. Notable tools in this space include ChatGPT, GitHub Copilot, Tabnine, Kite and Amazon Code Whisperer (Bull and Kharrufa 2024; Nguyen-Duc et al. 2023; Cheng et al. 2024). Numerous studies have discussed GenAI's impact on automating coding tasks, supporting developers and improving educational methods. GenAI has demonstrated its ability to enhance productivity, reduce human errors, automate code generation and documentation and find errors in software code. By streamlining these processes, GenAI can improve the quality of work and allow developers to focus on more complex, knowledge-intensive and creative aspects of their projects (Cheng et al. 2024; Coello et al. 2024; Kazuo 2024).

A key feature of GenAI is its ability to automate code generation. LLMs such as GPT-4 and tools like GitHub Copilot are integrated into development environments, supporting programmers with tasks such as writing, restructuring and testing code (Cheng et al. 2024; Coello et al. 2024; Komp-Leukkunen 2024; Nguyen-Duc et al. 2023). Most studies show that GenAI performs well on small, well-defined coding tasks and can produce high-quality code, but struggles with larger, more complex problems. Human oversight remains crucial to validate accuracy and ensure the code meets project requirements, highlighting the need for a balance between AI assistance and human judgement in the software development process (France 2024; Ulfsnes et al. 2024).

GenAI also plays a critical role in collaborative development. Ross et al. (2023) developed the 'Programme's Assistant', a conversational AI system that facilitates multi-turn discussions and code generation. Similarly, Ulfsnes et al. (2024) found that AI tools are changing team dynamics, as developers increasingly rely on AI for solutions instead of colleagues, which can disrupt traditional knowledge-sharing practices within agile teams. This shift highlights the dual-edged nature of AI integration: while productivity and efficiency increase, teamwork and collaborative learning might decrease, posing challenges to the long-term success of software development teams.

Furthermore, GenAI is an invaluable tool for training future software engineers. Bull and Kharrufa (2024) examine the role of AI-powered assistants in software development education, noting that AI tools can offer real-time feedback, assistance in finding programming errors and personalised learning experiences. This aligns with Komp-Leukkunen (2024), who discusses how tools like ChatGPT could democratise programming knowledge, enabling individuals with little formal education to acquire programming skills independently. This democratisation, while expanding the talent pool, may reduce the specialist status of software engineers, further complicating the job market for professionals in this field.

Like humans, GenAI is prone to making mistakes, underscoring the importance of quality control through reviews and

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testing. One effective strategy is leveraging GenAI to review tasks rather than generate design documents or code. This approach minimises the likelihood and impact of errors, as mistakes in review outputs tend to be less severe than errors in generated code. Furthermore, when GenAI is used to generate code, potential issues like hallucinations can be addressed more easily through automated checks, such as unit tests. Integrating GenAI into software development workflows introduces its own set of challenges. As noted by Clement et al. (2023), Cheng et al. (2024), Ulfsnes et al. (2024), and Kazuo (2024), the opaque nature of AI decision-making can hinder trust and understanding. Developers may need help interpreting or verifying AI-generated solutions, particularly for complex or mission-critical code. Explainable AI (XAI) models might be required to improve transparency, making it easier for developers to understand and modify AI outputs when necessary (Dwivedi et al. 2023).

GenAI reshapes software development by automating coding tasks, fostering collaboration and opening new educational opportunities. However, as Komp-Leukkunen (2024) and Ulfsnes et al. (2024) caution, integrating these tools requires careful attention to transparency, reliability, and a balanced role for human and AI contributions. This is essential to ensure GenAI enhances rather than undermines the creativity and productivity of software engineers. Although tools like ChatGPT can significantly boost productivity and lighten workloads, they raise concerns about job displacement, especially for routine tasks (Sauvola et al. 2024; Komp-Leukkunen 2024).

To sum up, GenAI is rapidly transforming the software development industry. In fact, due to the several advantages expected from using this new tool, this industry has been among the first and most intense adopters of such technology (Alavi et al. 2024; Ebert and Louridas 2023). Therefore, many scholars are interested in studying the impact of GenAI on the daily work of software developers. What currently needs to be added, however, is an analysis based on a KM perspective, which could offer relevant insights, given that software development is a knowledge-intensive activity. This is why we used the software developers as the 'application context' to conduct our empirical analysis.

2.3 | Analyzing the Impact of GenAI on KM Through the TOE Framework Lens

The previous sections have emphasised the importance of analysing GenAI's impact on organisational KM, especially in a software development context. Since we are still in the nascent stages of this phenomenon, exploratory studies remain crucial for identifying and delineating its main implications and potential. Nevertheless, referring to an established conceptual framework would help analyse and interpret such studies' findings more effectively. Among the various theories and frameworks used to study the adoption of new technologies (Gupta et al. 2024), we have chosen to employ the TOE framework developed by Tornatzky et al. (1990). This choice is based on the framework's previous practical application in studying the adoption of AI in general (Simina and Dutescu 2024; Yang et al. 2024) and also of GenAI (Agrawal 2024; Wael AL-khatib 2023; Neumann et al. 2024) and it was already used in the field of KM (e.g., Wang and Wang 2016). Moreover, this framework can be employed to investigate and analyse various factors due to its inherent malleability and flexibility (Maragno et al. 2023; Neumann et al. 2024). Such adaptability makes it well-suited for exploration without being constrained by predetermined constructs (Yang et al. 2024), which this study aims to do.

The TOE framework, developed by Tornatzky et al. (1990), is a comprehensive model used to examine how firms adopt and implement a new technology. It posits that three main factors influence an organisation's technology adoption decisions, as follows:

- Technological factors: refer to the characteristics of the new technology, including perceived benefits, compatibility, challenges and the accompanying infrastructure. These factors are crucial in adopting a new technology since they are key in determining the relative advantage offered and the challenges raised by the innovative tools. Some authors (Maragno et al. 2023; Yang et al. 2024) trace the technological factors back to technology affordances, which refer to what an individual or organisation with a particular purpose can do with a technology, and technology constraints, which refer to ways in which an individual or organisation can be held back from accomplishing a specific goal when using technology (Majchrzak and Markus 2012).
- Organisational factors: include the characteristics and resources of the internal organisational context, such as its size, managerial structure, working procedures and so forth. Organisational factors account for the compatibility of the organisation's structure and practices with the new technology. The ease with which the new technology can be integrated into daily routines is crucial in favouring its adoption and use (Neumann et al. 2024).
- Environmental factors: encompass the external environment in which an organisation operates, including industry requirements, competitive pressures, legal and regulatory environment and ethical considerations. Notably, emerging technologies increasingly emphasise aspects related to the protection and security of sensitive data and the validity of external data, reflecting growing concerns about privacy and data integrity in the digital age (Maragno et al. 2023).

It is worth recalling that the TOE factors can positively and negatively influence the adoption and use of a new technology. As previously mentioned regarding technological factors, the perceived relative advantages can accelerate the adoption and use of new technology, whereas its complexity may hinder the process.

3 | Research Methodology

3.1 | Research Design

Drawing inspiration from the KM-related issues raised by extant literature summarised in the previous section, this study investigated the purpose and use of GenAI during software

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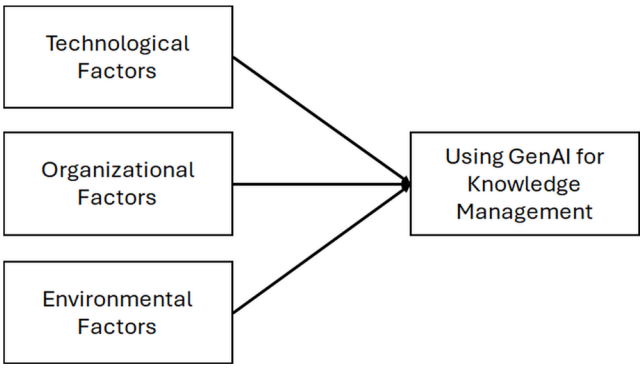

FIGURE 1 | Applying the TOE framework to investigate the GenAI influence on knowledge management.

development, which is considered a knowledge-intensive activity. It examined the main benefits and challenges connected with its use, the new competencies required for software developers using GenAI, and the impact it may have on interactions with colleagues and onboarding new hires. The findings were then interpreted from a KM perspective, focusing on the knowledge generation and sharing processes inherent in software development tasks. Then, we applied the TOE framework for technology adoption in organisations as a theoretical lens, with GenAI as a technology that influences the KM processes related to the software development activity (Figure 1).

Given the exploratory aim of the research, we adopted a qualitative approach. In particular, we applied a convenience sampling method. A convenience sample can be defined as a sample in which research participants are selected based on their ease of availability. Essentially, individuals who are the most ready, willing and able to participate in the study are the ones who are selected to participate (Given 2008). Convenience sampling is a popular method, especially when an exploratory study concerning an emergent phenomenon has to be conducted, in which initial insights are needed (Pham and Ho 2017; Zhang and Liu 2021), as in the case of GenAI. During February 2024, we conducted 11 semi-structured interviews with software developers from eight countries and with different levels of programming experience, aiming to capture a broad spectrum of experiences and insights. The semi-structured interviews covered topics like the usage of GenAI at the interviewee's workplace, especially for different tasks in software development, advantages and challenges of using this technology, and how the work and the interactions with colleagues are changing, for example, during the onboarding of new software developers. The semi-structured interview format allowed additional questions to dive deeper into some areas.

As no sensitive and person-identifying information was collected, our study did not require approval from an ethics committee. Before the interviews, all participants were informed about the reason for data collection, recording, and using their information for research purposes. Their informed verbal consent was obtained. This was the most convenient form of consent as all participants were interviewed online or face-to-face. The interviews were audio-recorded and subsequently transcribed verbatim. This transcription process enabled a thorough analysis of the responses.

We combined an inductive and a deductive approach for the analysis of the interviews. First, we followed an inductive approach for data analysis using Braun and Clarke's (2006) six-phase framework, allowing themes to emerge directly from interviews. Once themes were extracted, we applied a deductive approach by using the TOE framework as a theoretical lens to interpret and organise the findings. This mixed analysis allowed us to explore the data openly while also connecting the results to an established theoretical structure. This methodological approach is particularly effective for exploring complex, multifaceted topics, such as the role of GenAI as a KM support tool in software development processes.

3.2 | Profile of Interview Respondents

Table 1 summarises the profile of respondents, whose diverse experiences and backgrounds in software development and GenAI tools such as ChatGPT enrich insights and understanding of the impact of GenAI on software development practices and KM in different contexts. The interviewees span from self-employment and startups to employees at large corporations in the technology, finance, security and gaming sectors. They showcase a wide geographical and cultural diversity, influencing their software development and GenAI experiences. They hold various roles within software development, including technical leadership, coding and academic research, indicating a wide range of expertise and responsibilities and their experiences in software development range from short to very long.

4 | Results

As described in the methodology section, we employed a structured clustering approach to analyse the interview responses. This method enabled us to identify and organise the data into the following key themes: (1) usage and purposes for GenAI; (2) challenges with GenAI; (3) learning and adaptation; (4) current work practices integration; (5) public data and privacy. Following the TOE framework, we categorised the themes: the first two pertain to technological aspects, the next two to organisational aspects, and the final one to environmental aspects. Therefore, the empirical findings highlight that all three factors of the TOE framework are relevant in understanding how software developers adopt GenAI tools to support their KM-related tasks.

In the following sections, we present each theme in detail and discuss its implications from a KM perspective.

4.1 | Technological Affordances: Usage and Purposes for GenAI

This theme concerns the motivations behind incorporating GenAI into software development, especially its advantages. These essentially concern the characteristics of the new technology and, therefore, go into the technology factors of

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TABLE 1 | Profile of respondents.

No	Country	Role	Age	Years in software development	Years of experience with GenAI	Business sector
1	Denmark	Senior developer	61	30	1	Software company
2	Italy	Full-stack developer	26	2	1	Consulting company
3	Italy	Startup owner, developer	27	4	1.5	Startup, IT service
4	Denmark	Junior developer	28	6	1	Research institute
5	Germany	Junior developer	33	0.5	0.5	Press agency
6	Iran	Technical team leader	35	8	2	Financial services
7	UK	Software engineer	27	3	0.5	Software company
8	Sweden	Lead software engineer	34	10	0	Gaming industry
9	Iran	Mid-level developer	36	14	2	Financial services
10	France	Senior developer	52	28	0	Software company
11	The Netherlands	Senior developer	40	25	1	Software company

the TOE framework. The interviewed software developers increasingly rely on GenAI tools for various knowledge-intensive tasks, including debugging (discovering and fixing errors in software), code generation (automatic production of program code), code optimisation (changing a software code to make the software more efficient), and code refactoring (restructuring the code to improve readability, reusability and maintainability of a software code without changing code functionality). This confirms the malleable nature of GenAI technology. Its adoption aims to enhance the speed and quality of the development process, directly impacting KM by accelerating knowledge creation and application within projects. By facilitating rapid prototyping (creating an incomplete version of (a part of) software for early testing and customer feedback) and offering solutions to coding challenges, GenAI tools have become integral in the knowledge creation phase of KM, enhancing the productivity and creativity of developers. Some interviewees indicated that they use the tool to handle unfamiliar algorithms and for refactoring code. Others noted that by precisely describing their problem, they can receive a proposed solution. This shift underscores a move towards more efficient and automated KM, where knowledge is stored or retrieved and actively generated and applied in real-time to solve complex problems. GenAI tools reduce the time spent on repetitive tasks and foster an environment where high-quality knowledge assets are created and shared more efficiently among team members.

4.2 | Technological Constraints: Challenges With GenAI

This theme addresses the obstacles and issues when integrating GenAI-generated code into software projects. As they are related to the technological features of the new tools, this theme can be included among the technology factors of the TOE framework. According to our interviewees, developers may encounter problems such as code inaccuracies, a lack of context understanding by the GenAI, and difficulties debugging GenAI-generated code. There are concerns about the reliability and security of the code produced, as well as potential biases inherited from the training data. Managing the integration of GenAI outputs with existing codebases can be challenging, requiring additional validation and testing efforts. There is also the risk of over-reliance on GenAI tools, which might hinder the development of developers' coding skills and critical thinking abilities.

Although the benefits are clear, challenges such as ensuring the relevance, accuracy, and reliability of GenAI-generated code and integrating these solutions into existing projects highlight the need for robust KM practices, as a respondent expressed that they do not fully trust the tool and feel the need to verify its outputs. They use it as an initial proposal, which they then review and often need to modify. GenAI significantly influences the knowledge sources utilised by software

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developers. In particular, GenAI reduces the reliance on traditional sources such as forums (e.g., Stack Overflow, a website where computer programmers can ask questions and others can answer), blogs and consultations with colleagues. This shift is primarily due to GenAI's ability to offer rapid and straightforward solutions, eliminate the need for consulting multiple sources, and facilitate the interactive reformulation of questions. However, despite these advantages, GenAI is considered less reliable than traditional knowledge sources, especially for addressing uncommon, very specific or new problems.

GenAI might not be a solution for people new to programming. For example, another interviewee underlined that reviewing the code provided as a solution requires a high level of expertise to determine whether it is correct. These challenges call for a structured approach to managing the knowledge assets generated by GenAI, ensuring that they are accurately validated and effectively integrated into the collective knowledge base of the organisation. Also, passive use of GenAI can hinder the acquisition of new knowledge. This emphasises the need for critical engagement with GenAI outputs to ensure accuracy and relevance.

GenAI can quickly handle simple tasks typically assigned to inexperienced hires. However, this creates a dilemma: Traditionally, individuals gain experience by starting with these more straightforward tasks. This situation poses a critical question: How can workers become experienced if they cannot train on basic tasks?

4.3 | Organisational Factors: Learning and Adaptation With GenAI

This theme relates to how adopting GenAI influences the learning curve and adaptation processes within software development teams. It explores the impact on software documentation practices, as AI tools might change how documentation is generated and maintained. The introduction of GenAI necessitates developers to acquire new competencies and knowledge, such as understanding AI-generated suggestions and effectively integrating them into their workflow. It also affects job roles by shifting responsibilities and requiring continuous learning to keep up with AI advancements. Additionally, the theme deals with how GenAI alters interactions among colleagues, potentially enhancing collaboration through shared AI tools, creating challenges in communication due to varying levels of AI proficiency. For all these reasons, it can be traced back to the organisational factors of the TOE framework.

The interviews highlighted the impact of GenAI on learning and adaptation and its role in the continuous evolution of KM practices. GenAI requires users to ask accurate questions and clearly define problems, necessitating new competencies (knowledge) in question formulation. A respondent noted that users must be exact in how they formulate questions. This necessity has led to the emergence of a new discipline known as prompt engineering. Although few respondents said that they use GenAI tools for simple tasks, as developers engage with GenAI tools, they acquire new technical skills and develop sophisticated problem-solving abilities. This dynamic learning environment fosters a culture of continuous improvement and adaptation in KM processes, where knowledge is constantly updated, enriched and contextualised to meet emerging challenges. A critical question that emerged during the interviews is whether GenAI can be considered a 'new' team member. Some respondents noted that using GenAI leads to fewer interruptions when asking colleagues for help, streamlining specific processes. However, our observations suggest that this also results in less interaction between colleagues, reducing opportunities for one question to spark another and potentially limiting the creation and sharing of new knowledge. This raises important considerations about how adopting GenAI tools affects team learning dynamics. While GenAI can enhance efficiency, it also calls for re-evaluating how team members learn from one another and adapt their knowledge-sharing practices to maintain a culture of innovation and mutual education.

4.4 | Organisational Factors: Current Work Practices Integration

This theme focuses on how GenAI is woven into the fabric of current software development processes and its broader implications on the development lifecycle. It discusses the integration of GenAI into daily practices, such as code reviews, testing and deployment pipelines. It highlights how GenAI affects the onboarding of new hires, who may need training in effectively using AI tools. The integration of GenAI has the potential to redefine team dynamics and project management strategies, aiming for more efficient and innovative software development outcomes. This is why it can be included among the organisational factors of the TOE framework.

The transformative effect of GenAI on software development practices has profound implications for KM. GenAI tools are integral in creating a knowledge-rich environment by improving efficiency, facilitating collaboration and enhancing product quality. They enable the seamless integration of new hires into projects, effectively managing tacit and explicit knowledge within teams. As derived from the interviews, one of the significant skills they seek when hiring people is general knowledge and the ability to utilise GenAI.

In the future, soft skills and creativity will be crucial, especially in software development, where innovative solutions are needed for complex problems. The ability to view issues from different angles, which is essential for generating new knowledge, will be vital. GenAI tools can aid by providing diverse perspectives and enhancing the environment for effective software development.

4.5 | Environmental Factors: Public Data and Privacy

The last theme concerns the data sources that GenAI tools rely on, focusing on whether they use public datasets, private company data, or if there is uncertainty about their training data. It delves into data privacy and security concerns, especially

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TABLE 2 | Summary of findings in light of knowledge management.

Theme	TOE factor	Insights	Significant interviewee quotes
GenAI usage and purposes	T	Enhances efficiency by automating tasks like debugging and code generation, thereby accelerating development processes and improving code quality. This transformation shifts KM focus toward real-time knowledge generation and application, facilitating rapid knowledge creation, sharing and boosting productivity.	'I use it for algorithms that I don't know about, as well as code refactoring'. '... you can exactly describe your problem, and then, you will get the proposal for the solution'.
GenAI challenges	T	Necessitates robust KM practices for validating and integrating GenAI-generated knowledge. Uncritical use of GenAI hinders the generation of really new knowledge.	'... because you have to review the code provided as a solution, so you have to know, you have to be really knowledgeable to say: Is it correct or not'. 'those who use these tools cannot develop their problem-solving ability, during time, ...'
Learning and adaptation	O	GenAI tools foster a dynamic learning environment, enhancing developers' technical skills and problem-solving abilities while supporting continuous KM improvement. However, they may also reduce peer interaction, limiting spontaneous knowledge sharing and collaborative learning.	'I would ask ChatGPT, for example, for questions on certain terms that I don't understand'. 'We interrupt each other less for asking some questions. For instance, if I develop something that my colleague doesn't get it, simply asks the AI to explain that part for him/her'.
Integration in current work practices	O	Transforms development practices, creating a knowledge-rich environment that enhances collaboration and product quality. Ensures efficient onboarding and integration of new hires into projects, leveraging tacit and explicit knowledge.	'... one of the major skills we seek is to have general knowledge and the ability to utilise GenAI (in hiring people)'. 'I think it could definitely be used for onboarding, where it extracts the most relevant information...'
Public data and privacy	E	Raises concerns about the sourcing and use of knowledge, requiring transparent KM policies to maintain data integrity and confidentiality.	'I'm not allowed to do, due to company policies and legal issues'. '... since there is not a clear regulatory at the moment and it's an ongoing discussion in ... what ChatGPT produces is ... plagiarism'.

when proprietary codebases might be exposed to external AI services. Since these elements relate to the external context, they can be included among the environmental factors of the TOE framework. Developers and organisations are wary of the potential for sensitive information leakage and the implications for intellectual property rights. This theme highlights the need for transparency in how GenAI tools handle data, compliance with data protection regulations, and the importance of establishing trust in AI systems used within software development.

The mixed awareness and concerns regarding using public versus private data by GenAI tools reflect a critical KM issue: the sourcing and utilisation of knowledge. Some respondents stated that their companies are not allowing them to use these kinds of tools with sensitive data from the company due to company policies and legal issues. This calls for transparent KM policies that clarify the origin, use and sharing of knowledge, ensuring that GenAI tools enhance rather than compromise the integrity and confidentiality of organisational knowledge. However, openly sharing code with ChatGPT was viewed positively when possible, provided the code is not confidential or proprietary (e.g., if it is intended for research purposes or open-source development). From a knowledge-sharing perspective, this is an interesting point.

As a summary of the interview analysis, Table 2 synthesises the key insights into how GenAI is reshaping KM practices in software development, emphasising the opportunities for enhanced efficiency, learning and ethical knowledge handling.

5 | Discussion

Our study offers empirical evidence on the impact of GenAI on software development practices, which we examined through the double lens of the TOE framework and the KM notions. In particular, we identified the potential effects of GenAI on software developers' knowledge-related activities. These effects can be traced back to the factors outlined in the TOE model. The

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subsequent discussion focuses on these effects, especially delving into the implications of KM.

Similar to other findings (e.g., Ulfsnes et al. 2024), our empirical study found that GenAI is helpful in many aspects of software development, especially in code generation, software testing and learning. Respondents were, however, also aware of the challenges that the GenAI use brings. Although Alavi et al. (2024) look at the challenges of GenAI on knowledge workers in general, our paper dives deeper into a specific kind of knowledge work—software development. Our interview partners experienced difficulties due to errors in the GenAI-generated knowledge (the software code). Furthermore, they revealed that they rely more on GenAI to get help with their coding issues than to reach out to colleagues (e.g., on internet forums), as GenAI is simply quicker in providing solutions. As Alavi et al. (2024) discussed, there is a risk that knowledge workers reduce their collaboration and exchange with other knowledge workers because GenAI takes this role. Although we can see this effect in our data, the investigated developers still reach out to human colleagues when the development problem is very specific, so GenAI can only help a little.

In contrast to Alavi et al. (2024), who wrote that there is a risk that junior knowledge workers have to sidestep, as GenAI will better perform their entry tasks, our study revealed a more nuanced view. On the one hand, our interviewees mentioned the need for fewer young developers as GenAI can overtake simple tasks. On the other hand, the active use of prompt engineering combined with a critical attitude will become a necessary skill. Furthermore, GenAI can help onboard and train young developers, as they can use the tool to get explanations of existing code quickly and more readily into a new developer environment. At the same time, they still need their general development knowledge to build on to understand the GenAI-generated code. As repetitive coding tasks can be automated with GenAI, developers can concentrate more on complex tasks.

Even if focused on software development activities, our analysis confirms that GenAI is assuming the role of a game changer in organisational KM (Sumbal et al. 2024): in fact, KM approaches and activities are required to change to seize the opportunities and face the challenges given by GenAI. For instance, the fact that the artificially generated knowledge may be irrelevant, if not inaccurate, calls for its continuous verification. This does not only involve the ex-post check of the produced content but also creating the conditions for obtaining high-quality, relevant and reliable content, which requires not completely abandoning the traditional knowledge sources, favouring the development of a critical mind, and avoiding a passive use of the artificially generated knowledge. This should mitigate the risk that Alavi et al. (2024) underline, which is that the organisation loses control over the knowledge generated by GenAI.

Furthermore, we found that developers might focus on their work only and use GenAI as a knowledge source instead of contacting direct colleagues or other developers in internet forums. Ulfsnes et al. (2024) discuss that this could lead to less job satisfaction and lower team performance, but could also lead to more interaction about good prompting for getting good coding results from GenAI. In line with Sarka et al. (2019), knowledge managers need to be active in understanding and capitalising on the potential of this new technology for knowledge sharing and creation. To counteract the possible decrease of knowledge sharing among colleagues because resorting to GenAI reduces employee interaction and collaboration, the organisation should rethink socialisation and knowledge-sharing practices, for example, by reinforcing the activities for the reciprocal exchange of information and ideas. This should mitigate the negative side effect that Retkowsky et al. (2024) mention, where the intense use of GenAI breaks the existing knowledge ties within the organisation.

6 | Conclusion

The present study aimed to deepen our understanding of the impact of GenAI on organisational KM, specifically on knowledge creation and knowledge-sharing processes. This is an entirely new research topic, where the interest of KM scholars and professionals is growing rapidly, driven by the appearance of the latest technological tools, such as ChatGPT. To achieve the objectives of the study, we conducted 11 interviews with software developers who used GenAI in their daily activities. Given the knowledge-intensive nature of their job and that they are among the first adopters and users of GenAI tools, software developers represent a particularly promising study area for our research. The study's findings shed light on the current usage, benefits and challenges of using GenAI for software code generation and the generation of new knowledge. In particular, the interviews revealed that GenAI can help developers solve simpler, repetitive tasks, saving them much time. However, the source for the generated code is unclear, and developers need to learn how to formulate their questions to the GenAI tool to receive good results. Furthermore, continuous resorting to ChatGPT, on the one hand, allows for the rapid discovery of the needed knowledge. On the other hand, it could lead to reduced connections with team members, thus causing a decrease in knowledge sharing.

From an academic perspective, this study expands the currently limited empirical evidence on using GenAI to support KM processes. Moreover, by employing the conceptual framework provided by the TOE model, the study allows the identification of specific technological, organisational and environmental factors related to KM processes that can be incorporated into that framework to further investigate the adoption of GenAI in knowledge-intensive tasks. Furthermore, as we find that AI-generated knowledge represents a new type of knowledge that impacts organisational KM, we recommend that future work investigates this knowledge type further, particularly examining how existing KM frameworks may need to be revised to integrate AI-generated knowledge.

From the managerial point of view, it highlights that integrating GenAI tools into software development practices presents opportunities and challenges for effectively managing relevant knowledge. In particular, as these technologies continue to evolve, so must the KM practices that support their adoption, ensuring that knowledge is not only efficiently created and shared

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but also ethically managed and protected. In this regard, the results of the interviews illustrated in Section 4 provide specific suggestions for practical implications, as follows:

a. Re-focus KM activities by considering the type of knowledge that GenAI systems can automatically manage while adopting transparent practices that safeguard the integrity and confidentiality of organisational knowledge. This requires AI literacy among knowledge workers and a clear organisational adoption strategy to help knowledge workers identify AI-generated outputs and understand how to use GenAI as a new method of knowledge creation. In this context, prompt engineering emerges as a new form of knowledge creation; verifying and documenting effective prompts can lead to more efficient and responsible use of GenAI tools.

b. Adopt validation procedures to ensure the reliability of GenAI-generated knowledge. Human experts should tag and review outputs before being integrated into knowledge repositories. Encouraging team discussions around AI-generated solutions can mitigate the risk of collaboration gaps and help revive tacit knowledge sharing. The quality of knowledge and human-curated review remain essential. Additionally, encouraging users to actively engage with GenAI by not only consuming but also explaining and refining its outputs can reduce the risks associated with passive use.

c. Encouraging active knowledge exchanges between team members to preserve a dynamic learning environment. As knowledge sources become increasingly AI-driven, the pace of access accelerates, but there is a risk of losing depth. A solution is combining AI-generated knowledge with human team input to develop a KM system that balances speed and richness.

d. Enhancing onboarding and training for new employees by integrating GenAI tools alongside traditional ways of onboarding and training. This dual approach helps prevent the bypassing of essential tacit knowledge and learning processes. Furthermore, the ability to effectively work with GenAI tools, including validating AI outputs and engaging in prompt design, should be recognised as a new and necessary skill set in modern knowledge work.

Our study faces limitations due to the early adoption phase of GenAI among software developers, limiting comprehensive insights into its benefits and impacts on organisational KM. Currently, GenAI's application is mainly personal, with minimal integration into team processes, reflecting its limited influence on collective KM practices at this stage. Another limitation stems from the reliance on interviewees' perceptions, which may introduce biases or overemphasise personal experiences. This is a common challenge in interview-based research, which we attempted to mitigate by including workers from diverse countries, holding various roles and having different seniority levels.

Regarding future research, the findings from this present study should first be subjected to empirical confirmation. This could be accomplished through surveys utilising the TOE model constructs, appropriately integrated with the factors identified in this analysis. Future research should also focus on the evolution of GenAI's role in KM as it moves from individual application to organisational use. This shift needs to be examined, particularly regarding how GenAI affects team dynamics, employees' mutual collaboration and organisational KM processes. Furthermore, investigating changes in knowledge worker roles and skills in response to GenAI integration offers a promising direction for further study. As GenAI is used more frequently, knowledge workers should adapt their skills to use the tools efficiently.

Concerning software developers, it will be especially interesting to discuss which skills young developers will need for their future workplace and how their roles will be transformed, as GenAI can quickly solve simpler programming tasks.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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